



Co-selection of antibiotic resistance genes, and mobile genetic elements in the presence of heavy metals in poultry farm environments

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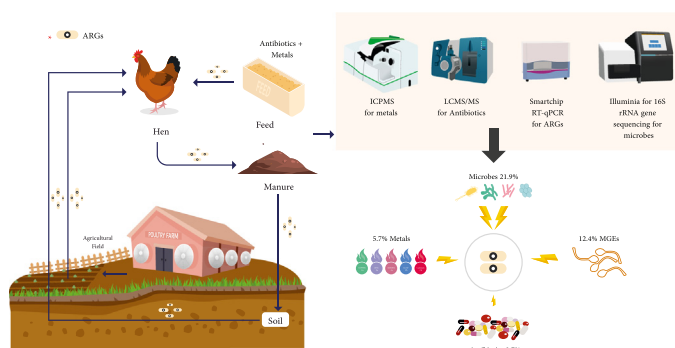
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HIGHLIGHTS

- Heavy metals in the environment trigger co-selection of antibiotic resistance in bacteria.
- MGEs facilitate the mobility of ARGs and MRGs.
- Bacterial composition and horizontal gene transfer were the major influencing factors on the composition of ARGs.
- Metals had a bigger effect on the ARG profile than detected antibiotics.
- Antibiotics, metals, MGEs and bacteria explain solely 0.7%, 5.7%, 12.4%, and 21.9% of variances in ARGs respectively.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 10 July 2020

Received in revised form 26 September 2020

Accepted 26 September 2020

Available online 2 October 2020

Editor: Ewa Korzeniewska

Keywords:

Co-selection

ARGs

Horizontal gene transfer

Heavy metals

Integrase

Transposons

ABSTRACT

Environmental selection of antibiotic resistance genes (ARGs) is considered to be caused by antibiotic or metal residues, frequently used in livestock. In this study we examined three commercial poultry farms to correlate the co-occurrence patterns of antibiotic and metal residues to the presence of ARGs. We quantified 283 ARGs, 12 mobile genetic elements (MGEs), 49 targeted antibiotics, 7 heavy metals and sequenced 16S rRNA genes. The abundance and type of ARG were significantly enriched in manure while soil harbored the most diverse bacterial community. Procrustes analysis displayed significant correlations between ARGs/MGEs and the microbiome. Cadmium (Cd), arsenic (As), zinc (Zn), copper (Cu) and lead (Pb) were responsible for a majority of positive correlations to ARGs when compared to antibiotics. Integrons and transposons co-occurred with ARGs corresponding to 9 classes of antibiotics, especially Class1 integrase *int1-1LC*. Redundancy analysis (RDA) and Variance partitioning analysis (VPA) showed that antibiotics, metals, MGEs and bacteria explain solely 0.7%, 5.7%, 12.4%, and 21.9% of variances of ARGs in the microbial community, respectively. These results suggested that bacterial composition and horizontal gene transfer were the major factors shaping the composition of ARGs; Metals had a bigger effect on ARG profile than detected antibiotics in this study.

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1. Introduction

Antibiotics are drugs commonly used around the globe, but now antibiotic resistant genes have become an increasing health risk (CDC, 2013; House, 2015; WHO, 2014). Thousands of tons of antibiotics are produced each year (Van Boeckel et al., 2015). Excessive use of antibiotics facilitated the spread of antibiotic resistance determinants in the environment (Wright and Poinar, 2012). Centers for Disease Control and Prevention (CDC) recorded around 2000 thousand morbidities and 23 thousand mortalities annually, due to antibiotic resistant bacteria, in the United States alone (House, 2015). The spread of ARGs is an emerging public health problem, and ARGs are now considered as an environmental pollutant (Stoll et al., 2012; Xiong et al., 2014). Actions are being taken by the World Organization for Animal Health in association with the World Health Organization (WHO) and the Food and Drug Administration (FDA) (Gilbert, 2012) for the regulation of ARGs. China has the largest industry of antibiotic production in the world. A study showed that the annual production of antibiotics amounted to approximately 210 million kg in China in 2007, with 46.1% of this share being produced for the livestock sector only (Hvistendahl, 2012), which is four times more than the amount that was being used in US livestock in 1999 (Joy et al., 2013). Manure is responsible for environmental pollution by spreading residual antibiotics (Alcock et al., 1999).

Antibiotics alone are not solely responsible for the development and dissemination of antibiotic resistance determinants; instead, a number of factors were shown to be involved (Andersson and Hughes, 2010). Moreover, metals mixed into animal feed is used to obtain fast growth and disease control due to increased immunomodulation in the animals being fed, while metals may also provide a long-term selective pressure for ARGs (Baker-Austin et al., 2006; Song et al., 2017). Recently, a direct relationship was discovered between the presence of heavy metals and ARGs. It could be shown that the co-selection of ARGs due to the presence of heavy metals was a major cause for the observed increase in the abundance of ARGs (Stepanauskas et al., 2006) and as serving as a selective agent in their proliferation (Allen et al., 2010). Metal resistance genes (MRGs) and ARGs have been co-selected by a variety of mechanisms in the environment (Ji et al., 2012). Mobile genetic elements (MGEs) were often in the immediate vicinity of these ARGs thereby facilitating co-transfer to other microorganisms by horizontal gene transfer (HGT) (Allen et al., 2010). Plasmids were shown to play a key role in the spread of both ARGs and MRGs (Fang et al., 2016). ARGs and MRGs share common modes of action when looking at the issue of environmental metal stress working as a co-selection agent to ease the proliferation of ARGs (Li et al., 2017), polluting the environment with ARGs. The presence of metals such as arsenic (As), mercury (Hg), cadmium (Cd), copper (Cu), cobalt (Co), chromium (Cr), lead (Pb), nickel (Ni), and zinc (Zn) is commonly associated with the simultaneous presence of ARGs and MRGs (Seiler and Berendonk, 2012; Yazdankhah et al., 2014; Zhao et al., 2019). MRGs are another serious public health issue since they have the potential to make these microbes more virulent, and can co-transfer along with ARGs and be found in various habitats, including the human gut (Ma et al., 2016). Rapid genetic exchange may occur in bacterial communities under environmental stress bolstering bacterial defense mechanisms (Li et al., 2017).

In the present study, we aimed to examine co-selection of ARGs by the presence of heavy metals in the environment of the poultry industry, where some metals (loids) are being used as boosters of the immune system of the chicken to aid growth, prevent disease, optimize feed utilization efficiency and improve meat pigmentation (Fisher et al., 2015). We sampled from three large-scale commercial poultry farms in Fujian province, China. To quantify ARGs we used HT-qPCR (Looft et al., 2012) with validated primers targeting ARGs and MGEs. We also tested excessively used veterinary and human antibiotics and metals in this study to verify our hypothesis along with 16S rRNA bacterial community analysis. Spearman correlation analysis, Procrustes analysis, RDA analysis combining with variance partitioning analysis

were used to link ARGs with MGEs, heavy metals and antibiotics to test our hypothesis of co-selection in the presence of heavy metals.

2. Materials and methods

2.1. Sampling sites and sample collection

A total of thirty-five samples were collected in 2017 from 3 large-scale commercial poultry farms located in Fujian province China (26°19'01"N 119°11'34"E, 26°11'35"N 119°10'39"E, and 26°20'17"N 119°09'06"E). Three types of samples: chicken manure, chicken feed and soil were collected. The sampled laying hens were kept in a multi-story battery cages. All the sampling birds were at the laying stage of production, and chicken manure of those farms was applied in agro-nomic fields. Freshly dropped feces were collected from cages and feed was collected from feed offered in containers from different places on a farm. All the samples were collected on the same day thereby excluding the effect of weather and season on the collected samples. Soil and feed samples were collected in sterile sealed plastic bags, while manure was collected in sterile plastic bottles well capped. The basic information about the manure, feed and soil was listed in Table S1. Samples were carried on dry ice for transportation and stored at −80 °C before DNA extraction and further analysis.

2.2. DNA extraction

High-quality DNA was extracted from 0.5 g of soil and feed samples using FastDNA® Spin Kit for Soil (MP Biomedical, France) and 0.5 g chicken manure using PowerFecal® DNA Isolation Kit (QIAGEN Laboratories, Inc.) following the manufacturer's instructions. DNA extraction negative control (PBS) was included during the extraction. PowerFecal® DNA Isolation Kit was having Inhibitor Removal Technology® (IRT) to remove inhibitory substances such as polysaccharides, bile salts and heme-containing compounds which are often found in feces. Agarose gel electrophoresis and ultraviolet-absorbance (ND1000, Nanodrop, Thermo Fisher Scientific Inc.) were used to determine the quality and concentration of extracted DNA. DES solution (50 µl) was used to elute the total DNA and stored at −20 °C until use.

2.3. Sequencing and 16S rRNA community analysis

To investigate the bacterial community composition, amplification of the hypervariable V4-V5 region was performed using primer 515F: GTGCCAGCMGCGCGGTAA and 907R: CCGTCAATCTCTTGAGTTT (An et al., 2018; Su et al., 2017; Turner et al., 1999; Zhao et al., 2018). One PCR negative control (sterilized water) was included during amplification. The amplicon sequencing was performed using the Illumina HiSeq2500 PE250 platform after being purified and normalized. Ion Plus Fragment library was used for the analysis of sequence libraries and Qubit 2.0 Fluorometer (Thermo Scientific) accessed their quality. UCHIME algorithm (Edgar et al., 2011) was used to filter chimera and low-quality reads from the raw reads. Quantitative Insights Into Microbial Ecology (QIIME) (Version1.7.0) performed the processing and analysis of the generated clean reads (Su et al., 2015). The default method was used in QIIME to pick up the open-reference Operational Taxonomic Units (OTUs). UCLUST clustering was used to assign same OTUs to the sequences showing ≥97% similarity (Edgar, 2010). When compared to the samples under study, the read numbers of the DNA extraction negative control and PCR negative control were too low to be considered for the further analysis. The sequenced data of 16S rRNA was processed and analyzed by "R" ("R-Project", 2018), mainly the R-package "phyloseq" (McMurdie and Holmes, 2013). Analysis of variance was performed to access the Alpha diversity (Shannon and Chao1 index) by R function "ano" in the package "stats" ("R-Project", 2018). Multiple comparisons between all groups under study were performed by the function "TukeyHSD" in the package "stats". Rarefaction curves

were generated by the R function “rarecurve” in the package “vegan” (Dixon, 2003) to evaluate mainly the sequencing depth. Non-metric multidimensional scaling (NMDS) (R function “ordination”, R-package “phyloseq” (McMurdie and Holmes, 2013)) based on Bray – Curtis distance was used to ordinate Beta-diversity of samples.

2.4. Antibiotics and metals quantitation

Multiclass antibiotics were extracted from chicken manure, feed and soil samples using modified QuEChERS (quick, easy, cheap, effective, rugged and safe) method. The sediment samples were freeze dried, homogenized and processed, were used as blank for this analysis and these samples were previously obtained from a drinking water reservoir (Shidou Upper Reaches, Xiamen, China). Samples were hydrated with 2.5 ml of 0.1 M Na-EDTA Mclvain buffer (pH 4.0) and then extracted by 5 ml mixture of MeOH and MeCN (1:3 v/v). The liquid-liquid partitioning between organic and aqueous phase was separated and dSPE (dispersion solid phase extraction) cleanup was performed using 100 mg mixture of C18 and PSA (1:2 w/w) and 50 mg MgSO₄. The extracted samples were analyzed using LC-MS/MS (liquid chromatography tandem mass spectrometry) based on our previously published protocol (Rashid et al., 2020). A total of ten groups of antibiotics were targeted in this study, while six groups of antibiotics were detected and observed in our samples, comprising of Penicillin, Tetracycline, Macrolide, Fluoroquinolone, Quinolone, and Sulfonamides. All targeted antibiotics and parameters of LC-MS/MS in MRM (+ESI) and chromatographic performance of antibiotics was mentioned in Table S2.

Samples were oven dried and digested in the microwave oven system ETHOS UP (Milestone, Italy). For the digestion process, soil and feed test portions of about 0.2 g and litter about 0.1 g were weighed into clean and dried 120-ml Teflon-PFA microwave digestion vessels. HNO₃ + H₂O₂ digestion acid recipe was used by adding 7.5 ml of HNO₃ and 1.5 ml of H₂O₂. After digestion, each aliquot was diluted with MilliQ water to make a total of 30 ml and analyzed using NexION 300× ICP-MS (PerkinElmer, USA) for the analysis of 75As, 52Cr, 60Ni, 63Cu, 66Zn, 111Cd, and 208Pb isotopes and two procedure blanks were used for measurement according to the described protocol (Zhang et al., 2012).

2.5. High-throughput qPCR

High-throughput qPCR was carried out by the Wafergen SmartChip Real-time PCR system (Wafergen, Fremont, CA). In the present study, 296 primer sets were applied, comprising: 283 ARGs belonging to the antibiotics; 8 transposases; the class 1, 2 and 3 integron-integrase gene (*intl-1*, *intl-2*, *intl-3*); the clinical class 1 integron-integrase gene (*cintl-1*) and a housekeeping 16S rRNA gene; (Su et al., 2015; Zhao et al., 2018). Further information regarding 296 primer sets is mentioned in Table S3. Wafergen SmartChip qPCR software presented melting curves automatically. All HT-qPCR reactions with three technical repeats and three negative controls without adding DNA template were performed. The HT-qPCR data for each reaction well with linear correlation coefficient R² smaller than 0.99 were discarded. All technical triplicates with all positive results were only considered for further analysis. The relative gene copy number was measured by following the protocol already published (Looft et al., 2012); relative gene copy number = $10^{(31-CT)/(10/3)}$, here CT denotes the qPCR results and 31 denotes the cut-off point. The division of the relative gene copy number by the relative copy number of 16S rRNA provides normalized gene abundance. Absolute quantification of qPCR on a Roche 480 system was used to measure absolute 16S rRNA copy numbers. Each 20 µl qPCR reaction mixture consisted of 10 µl 2 × Light Cycle 480 SYBR Green I Master, 1 µg µl⁻¹ bovine serum albumin, 1 µM of each primer, 1 ng µl⁻¹ DNA as a template and 6 µl nuclease-free PCR-grade water. Enzyme activation at 95 °C for 10 min initiated the thermal cycle, then denaturation happened at 95 °C for 30 s in 40 cycles, while annealing took

30 s at 60 °C and finally the extension at 72 °C for 15 s completed one thermal cycle. To standardize the calculations, a fragment of a cloned and sequenced 16S rRNA gene (1.39×10^{10} copies per liter) incorporated into a plasmid control for the generation of calibration curves.

2.6. ARGs analysis and correlation analysis

The index used for the alpha diversity was ARG observed richness. R function “aov” in the package “stats” was used for statistical analysis of alpha-diversity. The beta-diversity of ARGs was ordinated by principal coordinate analysis (PCoA) (R function “ordinate”, R-package “phyloseq” (McMurdie and Holmes, 2013)) based on Bray-Curtis distance. Permutational multivariate analysis (R function “adonis”, R-package “vegan” (Dixon, 2003)) was performed to test the differences in beta-diversity between groups. Heatmap, barplot and donut chart were plotted by ggplot2 (Ginestet, 2011). Spearman correlation analysis built into the function “rcorr” in the package “Hmisc” (Frank and Harrell, 2020) was used to test the correlations between metals/antibiotics and resistome. FDR algorithm was used to adjust the *P* value for various comparisons (Benjamini and Hochberg, 1995). The correlation network was visualized by the software “Gephi v0.9.2” (Leonard, 2004).

2.7. Procrustes analysis

The associations between microbiome and resistome were calculated by the Procrustes analysis. The Bray-Cutis distance matrices for genus-level microbial matrix and ARG relative abundance matrix were calculated by R function “vegdist” in “vegan” package (Dixon, 2003). Each distance matrix was ordinated using PCoA ordination by R function “cmdscale” in “stats” (“R-Project”, 2018) package. R function “protest” in “vegan” package (Dixon, 2003) was used to calculate the symmetric Procrustes correlation coefficients *r*, the sum of squared distance, *P* value and Procrustes residuals. The Procrustes correlations were plotted by ggplot2.

2.8. Redundancy analysis (RDA) and variance partitioning analysis (VPA)

To check the effects of different environmental variables on ARG composition, we used redundancy analysis by R function “rda” in “vegan” package (Dixon, 2003). The obtained explained variation R² was adjusted by R function “RsquareAdj” in “vegan” package (Dixon, 2003). The statistical significances of adjusted R² and the variance explained by environmental variables was done by Permutation test using the R function “anova” in the “vegan” package (Dixon, 2003). RDA result was visualized by the R function “ggord” (Beck, 2017). Variance partitioning analysis (VPA) was done by the R function “varpart” in “vegan” package (Dixon, 2003).

3. Results

3.1. Distributions of ARGs, MGEs, heavy metals, and antibiotics

A total of 230 unique ARGs and MGEs (8 transposase genes, 2 Class I integron-integrase genes, 1 Class II and 1 Class III integron-integrase genes) were detected from 35 samples of manure, feed and soil. Average of 104, 127 and 104 genes were detected separately in feed, manure and soil samples, which may resist nearly the majority of the classes of antibiotics comprising of aminoglycoside, chloramphenicol, beta-lactams, macrolide-lincosamide-streptogramin B resistance genes (MLSB), multidrug, tetracycline, vancomycin, and sulfonamides and others (Fig. 1A). Targeted genes representing a main antibiotic resistance mechanism as ‘antibiotic deactivation’ in manure and feed samples, while the ‘efflux pump’ had a prominent effect on soil samples. ‘Cellular protection’ mechanism is another main resistance mechanism in manure samples. Aminoglycosides were abundant as compared to other classes in the manure and feed samples, and multidrug resistance was much higher

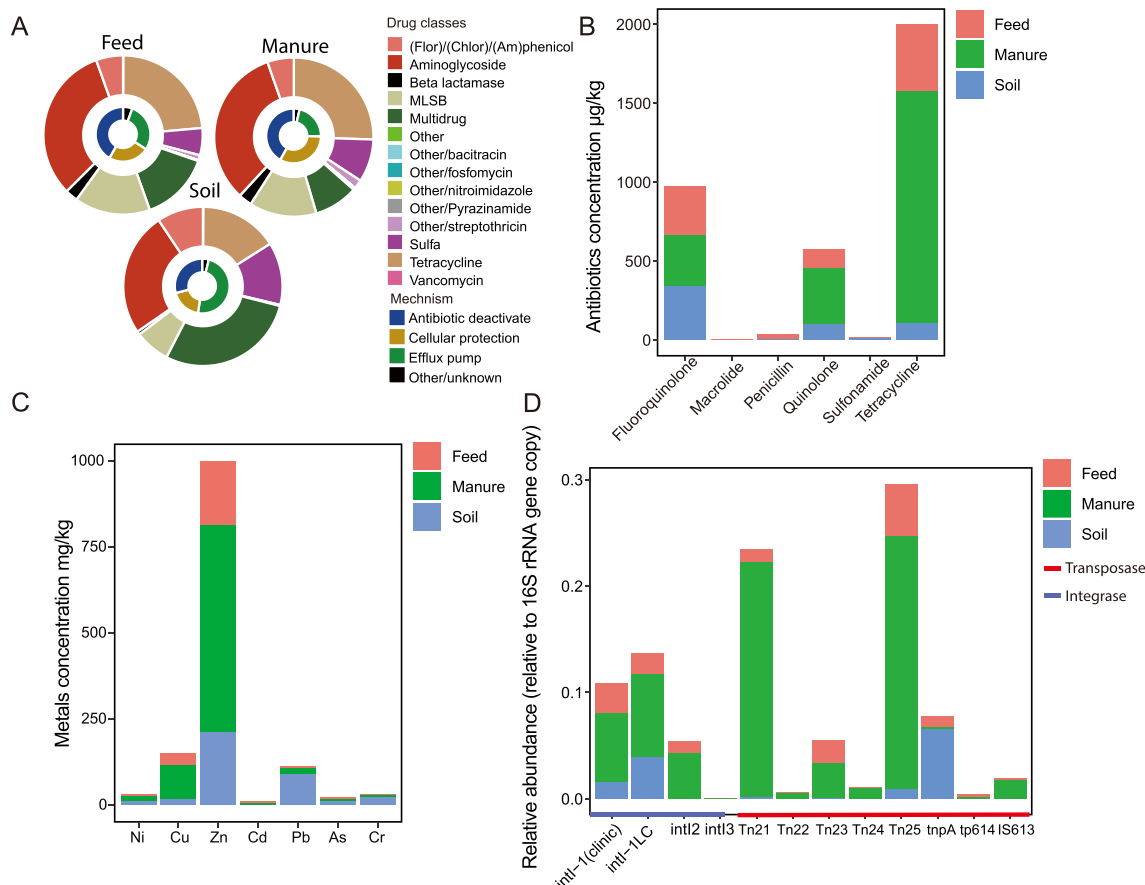


Fig. 1. Distributions of ARGs, MGEs, heavy metals, and antibiotics. (A) Drug classes and resistance mechanisms of ARGs detected in the samples. (B) The concentrations of major classes of antibiotics detected in the samples. (C) The concentrations of major metals detected in the samples. (D) The normalized relative abundances of MGEs in the three types of samples.

in soil as compared to other drugs (Fig. 1A). Overall, Aminoglycosides, tetracycline, sulfonamide drugs, multidrug and MLSB resistance genes were abundant in all samples while phenicol and beta-lactam were also considerable (Fig. 1A). Tetracycline, fluoroquinolone, quinolone were abundant in the three sampling sites, especially tetracycline in the manure (489.78 µg/kg) (Fig. 1B). Besides, Penicillin, sulfonamides and macrolides were not observed in manure samples while the concentration of penicillin was found higher (25.99 µg/kg) in feed samples (Fig. 1B). As for metals, zinc is the highest abundant metal among 7 detected metals in all the samples, especially in manure samples (544.4 mg/kg) (Fig. 1C). Followed by copper, lead was also detected to display a relatively high concentration in poultry farm soil (79.91 mg/kg). MGEs in manure were abundant compared to soil and feed samples, especially for two transposase genes *Tn25* and *Tn21*, two class I integron-integrase genes *int1-1*(clinical) and *int1-1LC*, one class II integron-integrase gene *int1-2* (Fig. 1D). One transposase gene, *tnpA*, was abundant in soil.

3.2. Enrichment of antibiotic resistance genes in manure

Manure had a significantly distinct ARG profile from the soil and feed samples shown in PCoA plot based on Bray-Cutis distance ($P < 0.01$, Fig. 2A). Specifically, manure had higher abundance (Fig. 2B) and higher richness (Fig. 2C) of ARGs. Similar results can be observed in the heatmap of ARGs as well (Fig. 2D). The ARG heatmap demonstrated all those ARGs with an abundance higher than 0.05 in the three groups. Aminoglycoside resistance genes, specifically *ant3ia*, *aph3ia*, *aph3iia*, and *aphi6a* were most abundant in manure and feed while multidrug resistance gene *qacEdelta* was consistently high in every environment irrespective of place or sample type. Among tetracycline resistance

genes, *tetQ* reached maximum relative abundance followed by *tetG*, *tetX*, *tetA*, and *tetM* in manure and feed. MLSB resistance genes *ermF*, *ermB* and sulfonamide resistant gene *sul2* were also highly abundant in all poultry farms under study.

3.3. Soil displayed a unique microbial composition

To measure the absolute abundance of bacteria in the samples, bacterial 16S rRNA gene copy number was measured by qPCR. Besides, we sequenced 16S rRNA genes in all the samples. A total of 3,048,521 clean reads were obtained, with an average length of 373 bp each read and clustered into 5627 OTUs at 97% similarity level. According to the adonis test, soil, feed and manure had individually distinct bacterial profiles visualized in the non-metric multidimensional scaling (NMDS) based on Bray-Curtis distance (Fig. 3A). Rarefaction plots (Fig. 3B) showed that soil bacteria were more diverse than manure and feed, and sequencing depths for feed and manure were enough for further analysis. In addition, microbial α -diversity using Chao 1 and Shannon index (Fig. 3C) showed that soil had significantly higher bacterial diversity across different sampling types, followed by manure. Taxonomic composition analysis also showed that the microbial community was more diverse in soil samples while Proteobacteria, Bacteroidetes and Firmicutes took up the majority of bacterial composition in manure samples, and Proteobacteria, Cyanobacteria in feed samples (Fig. 3D).

3.4. Co-occurrence of antibiotics, metals, MGEs and ARGs

Spearman correlation analysis was used to predict co-occurrences between ARGs and metals/antibiotics in this study. We employed a strict threshold (adjusted $P < 0.05$ & correlation coefficient $r > 0.7$) to

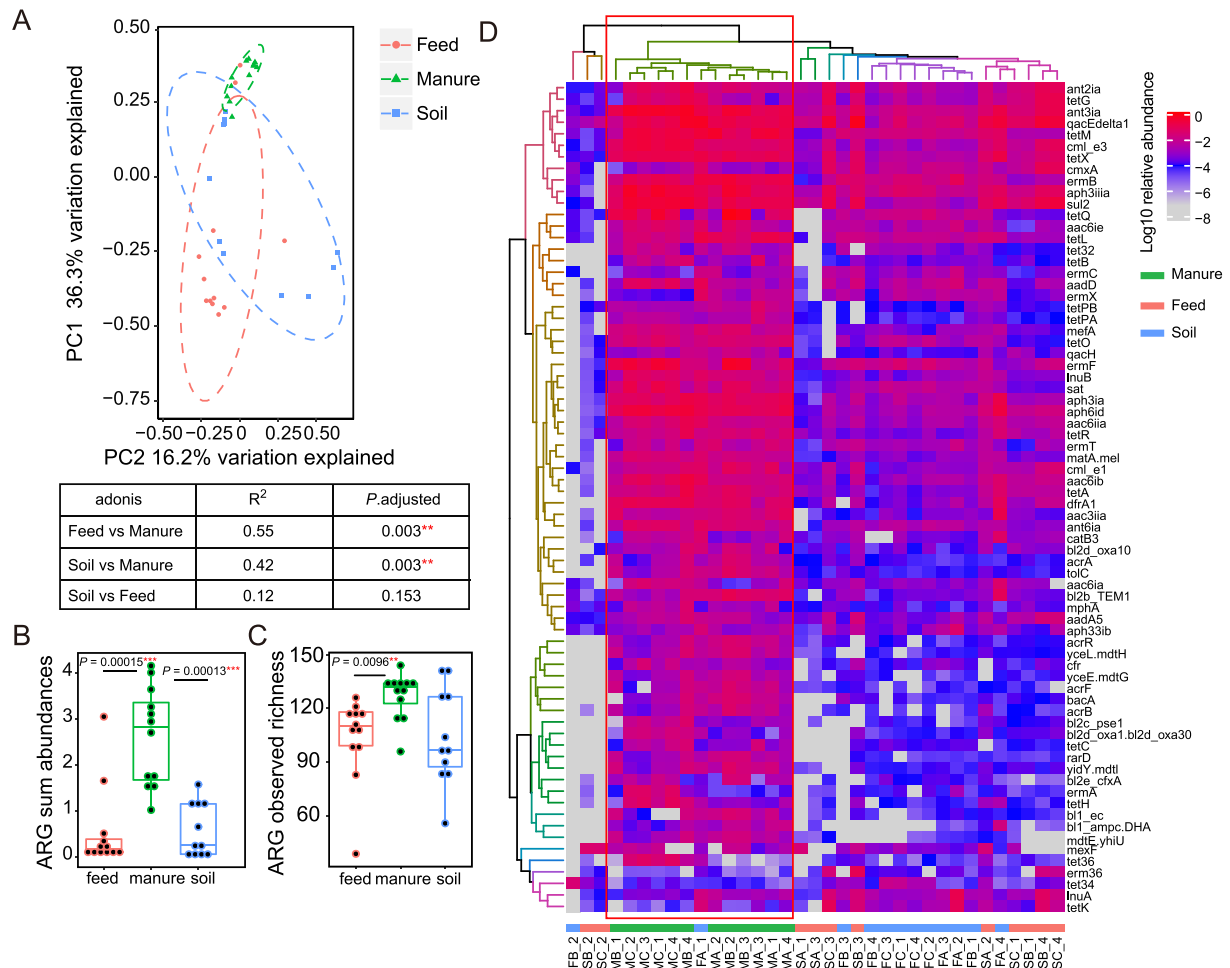


Fig. 2. Enrichment of antibiotic resistance genes in manure. (A) Principle coordinate analysis (PCoA) and adonis analysis based on Bray-curtis distance of ARG abundances from soil, manure and feed samples. (B) ARG sum abundances of soil, manure and feed samples. (C) ARG richness of soil, manure and feed samples. (D) Heatmap showing the abundances of all those ARGs with an abundance higher than 0.05 in manure, feed and soil samples from the three poultry farms. Red frame indicates the ARG abundance in manure. M, F and S represent manure, feed, soil, respectively and A, B, C indicates three poultry farms. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

define the significant correlations shown in the Fig. 4 and only those genes present at the least half the samples in each group was included in the correlation analysis. The network consisted of 5 heavy metals, 4 antibiotics and 27 ARGs (Fig. 4A). We did not detect any ARG which had correlations both with metals and antibiotics. Five metals including cadmium, arsenic, zinc, copper and lead accounted for a majority of positive correlations with ARGs, and antibiotics were not significantly correlated to ARGs compared to metals. Three major modules were observed with cadmium, arsenic and quinolones represented as hubs connecting to ARGs with significant correlations. Potential co-selection of cadmium was more diverse and stronger towards ARGs: tetracycline resistance genes, vancomycin resistance genes, beta-lactamase resistance genes, multidrug resistance genes, MLSB, and (flor)/(chlor)/(am)phenicol resistance genes. Three negative correlations were observed among cadmium with *msrA*, copper with *bl2be_shv2/2bl2_len* and fluoroquinolones with *mefA*. Arsenic co-selecting three tetracycline resistance genes (*tet36*, *tetH*, and *tetT*) one MLSB resistance gene (*erm35*), one bacitracin resistance genes (*bacA*), and one aminoglycoside resistance gene (*aac6ia*). Quinolone made a third large module links to MLSB resistance genes, vancomycin resistance genes, aminoglycoside resistance genes, and (flor)/(chlor)/(am)phenicol resistance genes.

A lot of significant correlations between MGEs and ARGs were detected in the samples as shown in the heatmap, delineating the significant impact of MGEs on ARGs (Fig. 4B). Nine classes of ARGs were

detected in this network to correlate with MGEs, of which aminoglycoside resistance genes, tetracycline resistance genes, and MLSB resistance genes were the most detected ones. Integrons and transposons (MGEs) co-occurring with ARGs corresponded to every class of antibiotics investigated in this study. Class1 intergrase *int1-ILC* was found influencing the most ARGs (46 interactions) compared to other integrases, next was the Insertion Sequence 613 (*IS613*) correlated with 40 ARGs and *Tn24* correlating with 36 ARGs in all the samples. *Tn24*, *Int1-2*, and *Tn22* were observed having a strong correlation with ARGs while *Tn22* was negatively correlated with vancomycin resistance.

3.5. Correlation between compositions of bacteriome to the composition of resistome

To investigate what extent to the bacterial composition dictates the resistomes (ARGs/MGEs), we performed Procrustes correlation which can be used to check the correlation degree of two ordinations (Peres-Neto and Jackson, 2001). This analysis confirmed significantly high correlations between ARGs (Fig. 5A, sum of squares $M^2 = 0.3086$, correlation $r = 0.8315$, $P = 0.001$)/MGEs (Fig. 5C, sum of squares $M^2 = 0.3339$, correlation $r = 0.8162$, $P = 0.001$) with the bacterial composition. These suggested that samples with similar bacterial compositions indicate similar resistome compositions. The between-sample type differences in resistomes seem to be partly explained by between-sample type differences in bacteriomes. In addition, we observed that

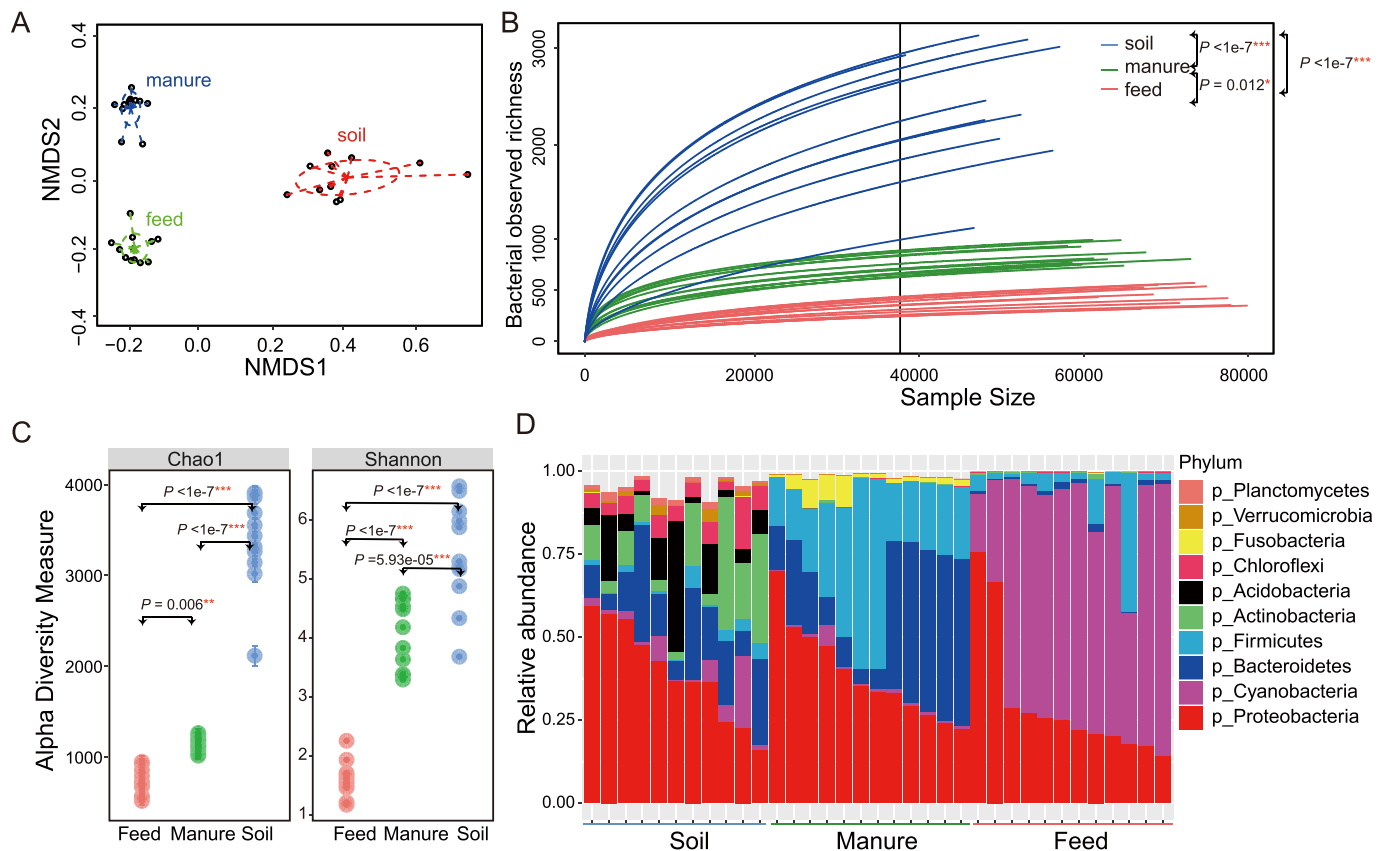


Fig. 3. The features of microbial compositions in soil, mature and feed samples. (A) NMDS ordination (stress value = 0.07) based on Bray-Curtis distance. (B) Rarefaction curves of samples based on bacterial observed richness (C) Alpha diversity of microbial community measured using Chao1 and Shannon index. (D) Relative abundances of the dominant bacterial phyla across samples in the soil, manure and feed. M, F and S represent manure, feed, soil, respectively.

the resistomes including ARGs (Fig. 5B) and MGEs (Fig. 5D) and bacteriome ordinations had more inter-individual variances in feed samples but ordinations in soil and manure samples yielded similar inter-individual results.

3.6. Influence of antibiotics, metals, MGEs, bacteria on composition and abundance of ARGs

To quantify what extent to the effects of covariates had on the abundance of ARGs, we performed redundancy analysis (RDA) which can demonstrate how the proportion of the community variations can be explained by external environmental variables (van den Wollenberg, 1977). In Fig. 6A, we investigated the environmental variables including 8 main bacterial phyla (85%–99.97%), antibiotics, metals, and MGEs. We also tested the effects of the 22 most abundant bacterial genera on ARG distributions. Genus *Brachyбактерium*, *Lactobacillus*, *Bacteroides*, *Escherichia-Shigella*, *Acinetobacter* and 5 other unclassified genera significantly affected ARGs (Permutation test for RDA, $P < 0.05$, Table S4). The amount of explanatory variation by all the environmental variables was 67.46% (adjusted R^2 , $P < 0.001$), which suggested that given environmental variables can explain the ARG composition well. Furthermore, the variances explained by antibiotics, metals, MGEs, Bacteroidetes, Actinobacteria, Proteobacteria were significant ($P = 0.001$ for antibiotics, metals and MGEs, $P = 0.001$ for Bacteroidetes, $P = 0.004$ for Actinobacteria, $P = 0.031$ for Proteobacteria) (Fig. 6A). Antibiotics, metals, MGEs and Bacteria can explain 10%, 31.5%, 39.6%, 45.8% of variances of ARG composition, respectively (Fig. 6B). After excluding the interactive effects between environmental variables, antibiotics, metals, MGEs and bacteria can explain solely 0.7%, 5.7%, 12.4%, 21.9% of variances of ARG community, respectively (Fig. 6B). These results suggested

that bacterial composition and horizontal gene transfer were the major influencing factors on ARG composition; Metals had a bigger effect on ARG profile than detected antibiotics in this study.

4. Discussion

Some previous studies have discussed the role of metals in co-selection of ARGs in manure (Song et al., 2017) or manure amended soil (Baker-Austin et al., 2006; Ji et al., 2012). To the best of our knowledge, no record of a complete study was available regarding the poultry environment, the biggest meat industry after pork (FAO, 2020) (FAO). This study displayed patterns co-occurrence involving ARGs and metals, antibiotics, MGEs, bacterial communities in poultry farms. Our results agree with previous studies illustrating that the key components influencing the ARG profile were the microbial composition (Xiong et al., 2014) and MGEs (Zhao et al., 2019). It could be further shown that the presence of metals had a greater impact on the ARG composition than the detected antibiotics (Li et al., 2017; Stepanauskas et al., 2006) and we extended these findings towards the poultry industry. It used to be a common practice to feed a complex mixture of antibiotics and metals to poultry birds for fast growth and immunity, which is now banned in most parts of the world including China. Currently, the use of mixtures of antibiotics and metals are prohibited in China for farming practices to avoid unfavorable results arising from their use. These measures are hoped to prevent the spread of ARGs in the environment and thus promote public health in the long term. We observed average concentrations of zinc, copper, tetracycline, and fluoroquinolones to be fairly high (Fig. 1B, C), and are suspected to select for ARGs in poultry gut microbiota. In addition, the mechanisms of resistance including antibiotic deactivation,

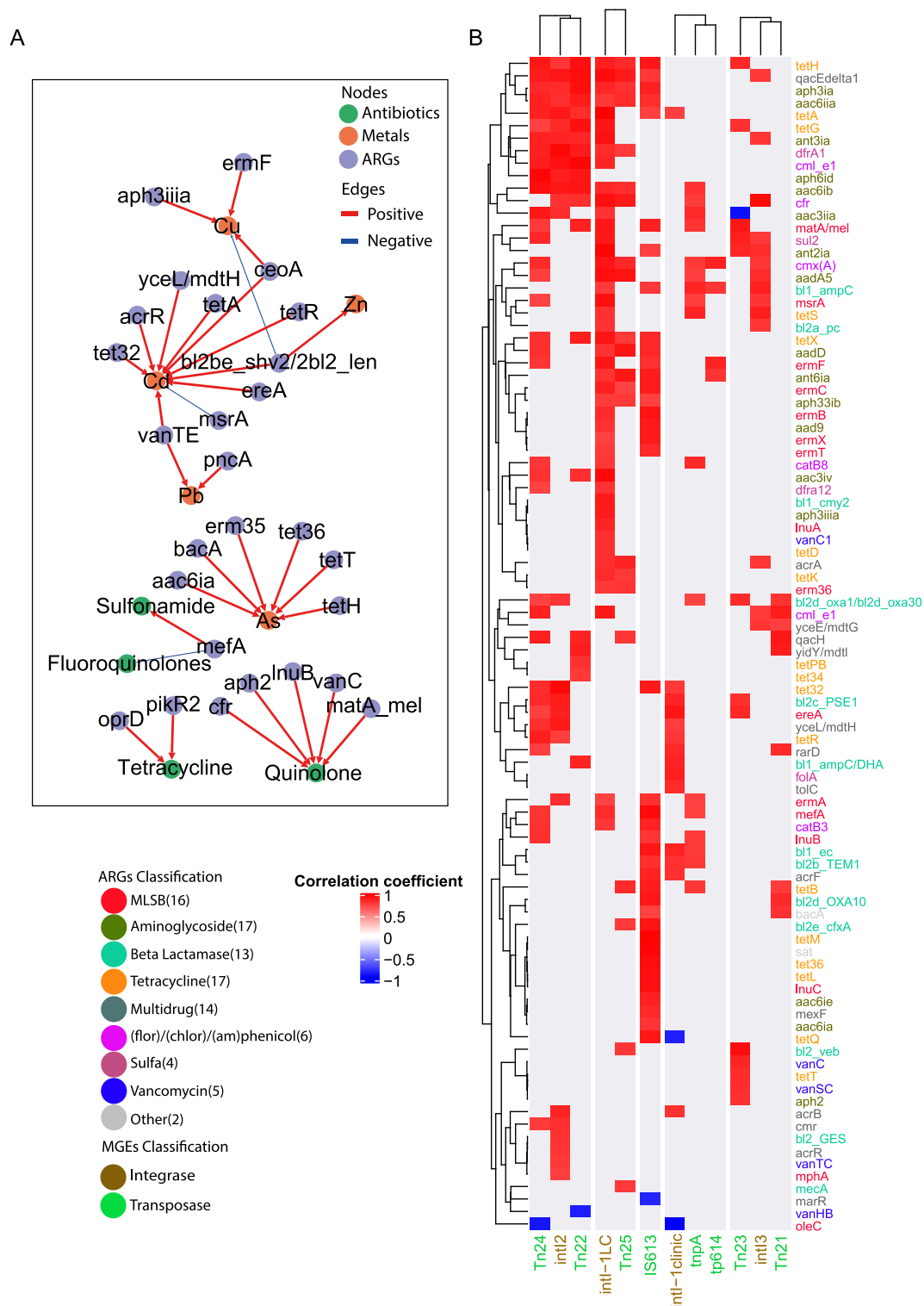


Fig. 4. Correlation analysis among metals, antibiotics, ARGs, MRGs and MGEs. (A) Network showing the significant correlations (adjusted $P < 0.05$ and correlation coefficient $r > 0.7$) among metals, antibiotics, ARGs and MRGs. Genes present in at least half the samples in each sample type were used for the correlation analysis. (B) Heatmap showing the correlation coefficients of the significant correlations (adjusted $P < 0.05$) between MGEs and ARGs in all the samples.

and efflux pumps that were detected to prevent cellular toxicity were consistent with previous studies (Zhao et al., 2018; Zhu et al., 2013). Strong enrichment of ARGs in manure samples was observed in Fig. 2A. According to previous studies, even a small dose of metal or antibiotic induced high levels of enrichment of ARGs in the gut microbiota, indicating that the internal microflora of birds might have played a pivotal role in the enrichment of antimicrobial resistance

(Osterblad et al., 1995). As a general phenomenon, sub-inhibitory exposure to drugs may enhance toxin and biofilm production (Rachid et al., 2000), thereby creating immunity in the gut microbiota by generating more potential for antibiotic tolerance by the phenomenon of co-selection (Goneau et al., 2015; Wong et al., 2000). Hence, we predicted ARGs to be enriched inside the gut of chicken, compared to external factors (feed, soil).

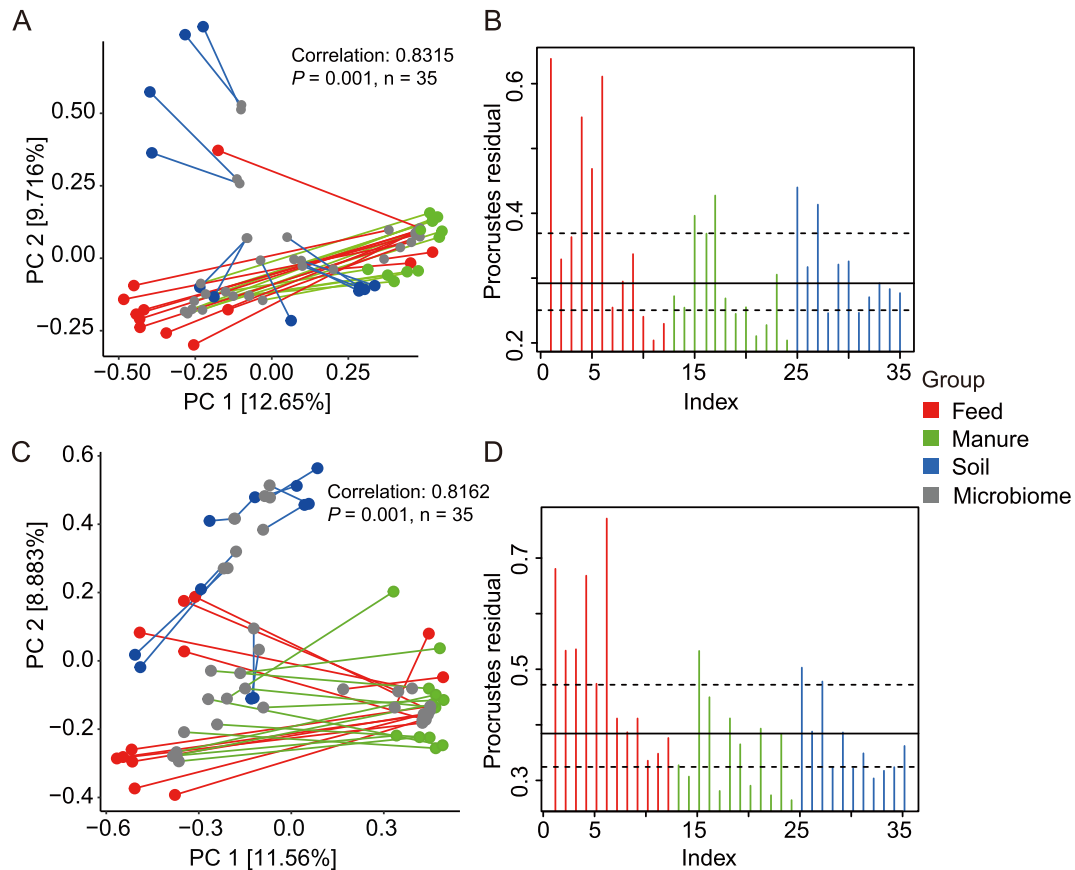


Fig. 5. Procrustes correlation analysis between microbial and ARGs/MGEs compositions. The correlations between microbial composition and ARGs (A)/MGEs (C) compositions, in the soil, manure and feed samples. The residual plots of Procrustes correlation analysis for ARGs vs bacteria (B) and MGEs vs bacteria (D) in feed, manure and soil samples; Horizontal lines symbolize the median (solid), 25% and 75% quantiles (dashed) of residual values.

Significant clustering was observed when examining the presence and abundance of metals and ARGs using Spearman correlation analysis (Fig. 4A). Non-degradable metals are more likely to persist in the environment and may accumulate to toxic levels compared to the residues of the antibiotics with high degradability. Thus, metals are more likely

to be critical in putting a long-term co-selection pressure on both clinically and environmentally important ARGs by genetic coupling in those contaminated settings (Li et al., 2017; Song et al., 2017).

The *cnr* and *czc* determinants shared extensive homologies to the encoded structural proteins as well as potentially exported cations

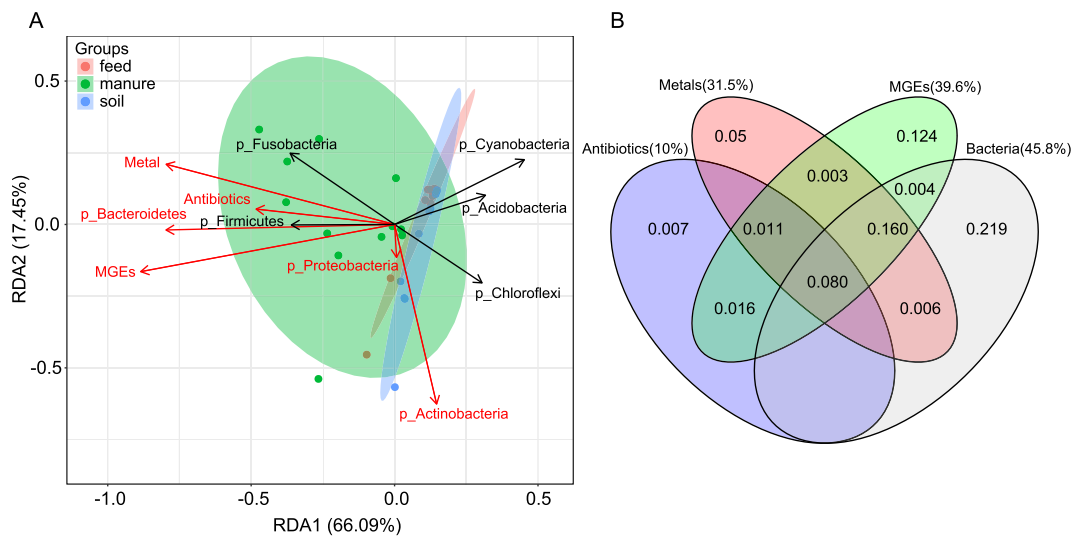


Fig. 6. Redundancy analysis (RDA) and variance partitioning analysis to demonstrate the effect of external environmental variables on the ARG community. (A) RDA plots of normalized ARG abundance and environmental variables. Environmental variables including eight main bacterial phyla, antibiotics, metals, and MGEs. (B) Venn diagram showing the variances of ARG compositions explained by antibiotics, metals, MGEs.

(Mazhar et al., 2020), whereas, arsenic, zinc, molybdenum, cobalt, silver, ciprofloxacin, tetracycline, chloramphenicol, and beta-lactam share some structural and functional properties using a reduced membrane permeability for cross-resistance (Baker-Austin et al., 2006; Ruiz et al., 2003; Silver and Phung, 1996). In the present study, we observed a significant clustering of ARGs with cadmium while zinc was in abundance, supporting the presence of mostly shared resistance determinants between zinc and cadmium (Fig. 1C). On the other hand, zinc showed less correlation even being most abundant probably because we employed a strict threshold (adjusted $P < 0.05$ & correlation coefficient $r > 0.7$) to define the significant correlations. Arsenic displayed a correlation to ARGs (Fig. 4A) especially, *tet* resistance determinants, that were likely enriched by co-selection. In contrast to metals, antibiotics did not significantly correlate to the presence of ARGs. These results indicate that poultry manure used in agricultural fields and the drainage water subsequently mixing with groundwater (Chen et al., 2017) would pose a public health risk.

Mobile genetic elements such as transposons, integrons and insertion sequences play a key role in the evolutionary process of specific members of the ARGs-containing microbiome, to help horizontal gene transfer of ARGs among microorganisms (Chen et al., 2017; Gaze et al., 2011; Gillings et al., 2015; Marshall and Levy, 2011). Class 1 integrase (*intI-1*) genes have been found to be abundant in bacteria in poultry litter irrespective of antibiotic usage (Nandi et al., 2004), while in the present study, class 1 integrase had the most significant correlations to ARGs resistant to multiple classes of antibiotics (Fig. 5). Studies have shown that class 1 integrons can capture various ARG cassettes mediated by an integrase (Gillings et al., 2015) while transposons are important in the propagation of these integron cassettes (Chen et al., 2017). The majority of integrases and transposases under the study were co-occurring with 9 antibiotics classes related ARGs, especially for tetracycline and aminoglycoside resistance genes (Fig. 4B) indicating that these ARGs might be transferred via HGT. A few vancomycin resistance genes were also observed to co-occur with MGEs which might be of great concern, as vancomycin is the last resort antibiotic against gram-positive bacteria.

The performed RDA and VPA quantified the importance of various environmental variables on ARGs. The selected environmental variables explained ARG composition well and a total of 67.46% variance of ARGs could be explained by them (Fig. 6A). Of them, metal, antibiotics, MGEs and 3 phyla including Bacteroidetes, Actinobacteria, and Proteobacteria exhibited significantly positive effects on the abundance of ARGs (Fig. 6A). Five known genera including *Brachy bacterium*, *Lactobacillus*, *Bacteroides*, *Escherichia-Shigella* and *Acinetobacter* were exhibiting significant effects on ARGs. The VPA showed microbial profile, MGEs, metals and antibiotics solely explained 21.9%, 12.4%, 5.7%, and 0.7% of variances of ARG community. MGEs have been fundamental to the evolutionary process and genetic variation of many living beings (Broaders et al., 2013). Quantification of the abundance of MGEs in microbial composition in an environment with selective pressure is important to estimate the influence of HGT on microbial evolution (Wright et al., 2008). The high abundance of MGEs (Fig. 1), their co-occurrence with ARGs (Fig. 4B) and the significant correlation between MGEs and bacterial community composition (Fig. 5) indicated the importance of MGEs in HGT in the current study. Followed by MGEs, heavy metals were also shown to have a great impact on the composition of ARGs (Figs. 4A & 6) even greater than the impact of detected antibiotics under study. Co-selection of the antibiotic resistance and the metal resistance in the microbiome is a serious issue as they may carry and disseminate ARGs, even without the presence of antibiotics (Pal et al., 2017, 2015). In the context of our results, genetic co-occurrences have suggested that MGEs facilitate the HGT of antibiotic and metal resistance determinants between bacteria, which provide opportunities for metals to co-select for antibiotic resistance across bacteria.

5. Conclusion

Our investigation on poultry farms suggested that metal pollution may have significantly affected the maintenance of a broad spectrum of ARGs. Previous literature addresses the issue in soil and pig farming; however, much less research has been performed on poultry farms. In this present study, we have studied how the abundance of ARGs is correlated with the abundances of MGEs, heavy metals, antibiotics and the microbial community in poultry farm samples. We have found evidence that heavy metals, used in poultry for growth promotion, and MGEs co-selected for ARGs, with MGEs facilitating the mobility of ARGs and MRGs. Directly or indirectly, poultry may thus play a pivotal role in public health by contributing to the spread of ARGs. Serious consideration may be required to mitigate the risk by researching metal-induced co-selection of antibiotic resistance genes in metal-rich environments for public health concerns.

CRedit authorship contribution statement

SHM, CR and RWF designed the experiments. YJW, SHM, JMS and JQX contributed in collection of samples from poultry farms. SHM, and AR performed this experiment. SHM and XJL wrote this manuscript. XJL and AD worked on the statistical analysis and graphical work. SGZ, JQS, YGZ, XJL, RWF and CR participated in the design of the study and revised the manuscript. All authors read and approved the final manuscript.

Declaration of competing interest

We declare that no conflict of interest exists in this manuscript, and manuscript is approved by all authors for publication. This manuscript is original and has not been published in whole or in part previously.

Acknowledgments

This research was funded by the National Natural Science Foundation of China (NSFC) grant number [31770123].

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2020.142702>.

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